

# SUMIT YADAV

Senior Software Engineer → Staff Engineer / Tech Lead

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## PROFESSIONAL SUMMARY

Senior Software Engineer and Tech Lead with **7+ years of experience** designing, building, and leading high-performance cloud-native applications at enterprise scale. Currently at **Accenture** leading a **team** and providing technical leadership across multiple full-stack and microservices workstreams involving **30+ engineers**. Deep expertise in **distributed systems, AI/RAG engineering, real-time architecture, and cloud infrastructure** across AWS and Azure. Author of a technical blog with 8 published deep-dives on system design and AI engineering. Actively targeting **Staff Engineer / Tech Lead** roles with cross-team technical impact.

## TECHNICAL SKILLS

### Frontend & Full-Stack Development

Node.js, TypeScript, React, Next.js, JavaScript, Express.js, NestJS, GraphQL, Tailwind CSS, HTML/CSS

### AI & Machine Learning

LLMs, RAG Pipelines, Vector Embeddings, Pinecone, Vercel AI SDK, Google Gemini 2.5, Prompt Engineering

### Databases & Caching

PostgreSQL, MongoDB, Redis (Upstash), DynamoDB, Pinecone (Vector DB)

### Cloud & Infrastructure

AWS (CloudFront, Lambda, DynamoDB, SQS, SNS, S3), Azure (Functions, App Service, DevOps, Storage), Vercel, Docker, Kubernetes, Nginx

### Architecture & Systems

Distributed Systems, Microservices, System Design, Event-Driven Architecture, Kafka, WebSockets, SSE, REST, GraphQL

### Leadership & Engineering Practices

Technical Leadership, Team Management, CI/CD, Code Reviews, Agile/Scrum, Performance Optimisation

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer (Tech Lead) · Accenture Solutions Private Limited

Mumbai, India | Dec 2021 – Present

- Led a team and provided mentorship and technical guidance to a team of 30+ developers across full-stack and microservices workstreams — conducting code reviews, defining engineering standards, and driving architectural decisions for enterprise-scale applications
- Designed and delivered Node.js backend modules and REST APIs for high-traffic enterprise applications, achieving 20–30% API performance improvement through targeted caching strategies and query optimisation
- Architected and implemented Azure-based cloud solutions including Azure Functions, App Service, and Storage Accounts, integrated into CI/CD workflows via Azure DevOps pipelines
- Defined microservice architecture patterns in collaboration with solution architects — establishing deployment workflows, service boundaries, and inter-service communication contracts
- Built production AI/RAG pipelines integrating LLMs, vector embeddings, and semantic search — delivering intelligent document Q&A systems on Next.js and Vercel
- Mentored engineers in scalable design patterns, cloud-native development, security best practices, and TypeScript — consistently raising team technical standards
- Recognition:** *Care Deeply For All Our People Award (Feb 2024) · Incredible Ownership Award (Sep 2023)*

### Senior Systems Engineer · Infosys Limited

Pune, India | Sep 2021 – Dec 2021

- Worked as a Full Stack Developer delivering features across JavaScript, Node.js, and modern frontend technologies in a fast-paced enterprise environment

## Software Engineer · [Accenture Solutions Private Limited](#)

Mumbai, India | Jul 2020 – Sep 2021

- Developed and maintained frontend web applications using React, JavaScript, and TypeScript — collaborating with backend teams on REST API integration and UI performance optimisation
- Contributed to component libraries, responsive design systems, and cross-browser compatibility initiatives across enterprise client projects

## Assistant System Engineer · [Tata Consultancy Services](#)

Mumbai, India | Dec 2018 – Dec 2019

- Worked as a Full Stack Web Developer building and maintaining web applications, contributing to both frontend and backend development using JavaScript and Node.js
- **Recognition:** *Kudos Award for Excellent Performance (Feb 2019)*

## ENGINEERING PROJECTS

### AlgoLens — Interactive Algorithm Visualizer · [Live Demo](#) · [Source Code](#)

Next.js 16 · TypeScript 6 · Vanilla CSS · React 19 · Vercel

- Built 25+ interactive step-by-step algorithm and system design visualizers from scratch — zero external animation libraries — covering sorting, searching, graphs, trees, dynamic programming, and distributed systems flows including LRU Cache and Consistent Hashing
- Architected a universal AnimationStep contract decoupling algorithm logic from the UI — every visualizer is a pure function producing steps consumed by a shared playback engine, making new algorithms addable in 3 files

### AI-Powered RAG Assistant · [Live Demo](#) · [Source Code](#)

Next.js 16 · Google Gemini 2.5 · Pinecone · Vercel AI SDK · TypeScript · Upstash Redis

- End-to-end RAG pipeline — PDF ingestion, text chunking, Gemini vector embeddings, Pinecone semantic search, and real-time streaming responses with source citations
- Production-hardened with embedding cache, rate limiting (Upstash sliding window), and mock dev mode to control API costs

### Real-Time Notification Engine · [Live Demo](#) · [Source Code](#)

Next.js 16 · SSE · Upstash Redis · QStash · TypeScript · Vercel Edge

- Event-driven notification system with SSE delivery, Redis persistence, and QStash fan-out across In-App, Email, and SMS channels with auto-reconnection logic

### Distributed URL Shortener · [Live Demo](#) · [Source Code](#)

Next.js · Upstash Redis · Vercel Edge Functions · TypeScript

- AP-architecture URL shortener with O(1) Redis lookups, configurable TTL expiry, and click analytics deployed on Vercel Edge Functions for global low-latency redirects

## TECHNICAL WRITING

Author of [sumityadav-dev.vercel.app](https://sumityadav-dev.vercel.app) — 8 published deep-dives on distributed systems, AI engineering, and Staff-level architecture thinking including: SAGA Pattern, RAG Pipelines, Consistent Hashing, Message Queues (SQS/Kafka/SNS), Scaling Social Feeds, Real-Time Systems and Interactive Algorithm Visualizer.

## EDUCATION

### B.Tech in Computer Science and Engineering · [Manipal University Jaipur](#)

Jaipur, India | 2014 – 2018 | CGPA: 8.03 / 10

Portfolio: [ysumit99.github.io](https://ysumit99.github.io) | Blog: [sumityadav-dev.vercel.app](https://sumityadav-dev.vercel.app) | Open to Staff Engineer / Tech Lead opportunities